

Joel Jacob

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Education

Carnegie Mellon University – BS in Mechanical Engineering

Expected May 2027

Skills

Design & Analysis: Onshape, SolidWorks, ANSYS (Discovery, Fluent), Mastercam

Controls & Programming: Python, MATLAB, Java, C++, Arduino, Raspberry Pi, Git

Manufacturing & Fabrication: 3D Printing, Machining, Laser Cutting, Soldering, Assembly

Projects

NASA Lunabotics Challenge

Nov 2025 – Present

- Designing, 3D printing, and machining excavation subsystem components as a member of CMU Moon Miners to compete in the NASA Lunabotics Challenge, focusing on efficient lunar regolith collection
- Devised a dual-channel bucket excavator design that outperforms the baseline in terms of material throughput
- Minimized weight through structural FEA in ANSYS Discovery while maintaining required safety factors

National Havoc Robot League

Sept 2024 – Present

- Founded a new robot project under CMU Combat Robotics, coordinating a team to design, develop, and manufacture a 3 lb undercutter combat robot to compete in the National Havoc Robot League
- Designed 10+ custom robot components on Onshape, including a TPU chassis and AR500 steel spinner weapon, reducing fabrication errors and accelerating iteration before machining and 3D printing
- Optimized ideal motor Kv and weapon moments of inertia using MATLAB analysis to maximize impact effectiveness while retaining stable rotational performance

FIRST Robotics Competition

Oct 2021 – June 2024

- Oversaw the 12-member control department as the director of FRC Team 5123 to architect, network, and program a complete CAN bus-based controller system to win the 2023 FRC NYC Regional Championship
- Implemented Java command-based robot code with custom driver controls, improving driving precision and on-field responsiveness during competition
- Improved autonomous driving accuracy and reduced steady-state error by 65% by implementing vision-based localization with Raspberry Pi and tuning PID gains

Experience

Research Assistant, Carnegie Mellon University – Pittsburgh, PA

Sept 2025 – Present

- Contributing to the development of a wearable assistive device as a member of the MetaMobility Lab at CMU by integrating sensing and deep learning to improve human mobility support
- Redesigning hip exoskeleton hardware for improved user fit by applying human biomechanics principles, leveraging SolidWorks and 3D printing to achieve precise motor-joint alignment and a 30% weight reduction

Retail Sales Associate, S & A Stores Inc – Bronx, NY

June 2025 – Aug 2025

- Maintained organization of products across high-traffic areas, integrating loss prevention protocols, to improve overall customer experience, comply with presentation standards, and reduce product stock-outs by 20%

Green Team Associate, Groundwork Hudson Valley – Yonkers, NY

June 2022 – June 2024

- Reduced harmful sources against native biota by taking part in environmental engineering at over 20 job sites